

Q1 A 1.75 kg mass moves as a function of time as follows: $x(t) = 4 \cos(1.33t + 0.2\pi)$
 (a) What is amplitude, frequency, ang. frequency, and period of this motion? (b) What is the equation of the velocity of this mass? (c) What is the equation of acceleration of this mass? (d) What is the spring constant?

Given $x(t) = 4 \cos(1.33t + 0.2\pi)$

(a)

Amplitude (x_0) = 4 m

Angular frequency (ω) = 1.33 rad/sec

Phase (ϕ) = $0.2\pi \rightarrow 0.628$ rad.

Frequency (f) = $\frac{\omega}{2\pi} = \frac{1.33}{2\pi} \rightarrow 0.2117$ Hz

Period (T) = $\frac{2\pi}{\omega} = \frac{2\pi}{1.33} \rightarrow 4.7218$ s

(b)

$v(t) = -A\omega \sin(\omega t + \phi)$

$v(t) = -4(1.33) \sin(1.33t + 0.2\pi)$

$v(t) = -5.32 \sin(1.33t + 0.2\pi)$ m/s

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(c) $a(t) = -A\omega^2 \cos(\omega t + \phi)$
 $a(t) = -4(1.33)^2 \cos(1.33t + 0.2\pi)$
 $a(t) = -7.0756 \cos(1.33t + 0.2\pi)$ m/s²

(d) Spring Constant $\rightarrow k = m\omega^2$

$k = (1.75)(1.7689)$

$k = 3.095$ N/m

Q2. A 100-g ball hangs from a meter-long string. It swings as shown in following figure at $t=0$ and it is moving to the right. Find the phase constant (ϕ) in degrees. Sketch the x vs t and v vs t graphs. When will the ball pass through equilibrium?

$x(t) = A \cos(\omega t + \phi)$

$A = 0.5$ m

At $t=0 \rightarrow -0.65 = 0.5 \cos(\phi)$

$\rightarrow \phi = 1.70$ rad

$x(t) = 0.5 \cos(3.13t - 1.70)$

$v(t) = -1.565 \sin(3.13t - 1.70)$

At $x=0$

$0 = 0.5 \cos(3.13t - 1.70) \rightarrow 3.13t - 1.70 = \pi/2$

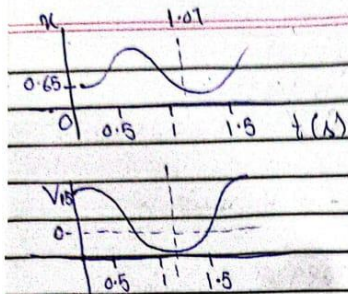
$t = 1.07$ s

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(3) a) $A(t) = A_0 e^{-\gamma t}$, $\gamma = \frac{b}{2m}$

At $t = 4T$

$$\frac{A(4T)}{A_0} = e^{-\gamma(4T)} = \frac{3}{4}$$

Take ln:

$$\gamma(4T) = -\ln\left(\frac{3}{4}\right) = -\ln(0.75)$$

$$\gamma = \frac{0.28}{4T}$$

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{10.0}{2.00}} = \sqrt{5} = 2.23606 \text{ rad/s}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{2.23} = 2.809 \text{ s}$$

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$$\gamma = \frac{0.28}{4(2.809)} \Rightarrow 0.0255 \text{ s}^{-1}$$

$$b = 2m\gamma \Rightarrow b = 2(2.00)(0.025)$$

$$\boxed{b = 0.102 \text{ kg/s}}$$

b) $E_0 = \frac{1}{2} k A_0^2 = \frac{1}{2} (10.0)(0.250)^2$

$$E_0 = 0.5 \times 10 \times 0.0625 = 0.3125 \text{ J}$$

After 4 oscillation amplitude:

$$A_f = \frac{3}{4} A_0 = 0.1875 \text{ m}$$

$$E_f = \frac{1}{2} k A_f^2 = \frac{1}{2} (10.0)(0.1875)^2$$

$$E_f = 0.17578 \text{ J}$$

Energy lost

$$\Delta E = E_0 - E_f = 0.3125 - 0.17578$$

$$\boxed{\Delta E = 0.137 \text{ J}}$$

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$$c) \omega = \sqrt{\frac{k}{m}} = \sqrt{5} = \boxed{2.2360 \text{ rad/s}}$$

$$\omega_d = \sqrt{\omega_0^2 - \gamma^2} = \sqrt{(2.236)^2 - (0.0256)^2}$$

$$\boxed{\omega_d = 2.235 \text{ rad/s}}$$

$$T = \frac{2\pi}{\omega_d} = \frac{2\pi}{2.2359} = \boxed{2.81 \text{ s}}$$

④ Given:

$$m = 7 \text{ kg} \quad T = 2.60 \text{ s}$$

$$T = 2\pi \sqrt{\frac{m}{k}}$$

$$k = \frac{4\pi^2 m}{T^2}$$

$$k = \frac{4\pi^2 (7.00)}{(2.60)^2}$$

$$\boxed{k = 40.9 \text{ N/m}}$$

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$$T = 2\pi \sqrt{\frac{L}{g}} \rightarrow g = \frac{4\pi^2 L}{T^2}$$

~~$$g = \frac{4\pi^2 L}{T^2}$$~~

$$\frac{L}{L_c} = \frac{g_T}{g_c} \quad \frac{g_T}{g_c} = \frac{0.9927}{0.9942}$$

$$\boxed{\frac{g_T}{g_c} = 0.9985}$$

$$⑥ \text{ Given } f = 1.50 \text{ Hz} \rightarrow T = \frac{1}{1.50}$$

$$T = 2\pi \sqrt{\frac{L}{g}} \rightarrow f = \frac{1}{2\pi} \sqrt{\frac{g}{L}}$$

$$L = \frac{g}{(2\pi f)^2}$$

$$L = \frac{9.8}{(2\pi \times 1.50)^2} \Rightarrow L = 0.1103 \text{ m}$$

$$\boxed{L = 0.11 \text{ m} = 11 \text{ cm}}$$

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① Given $f = 440 \text{ Hz}$, $A = 3 \text{ mm} = 0.003 \text{ m}$

$$\omega = 2\pi f = 2\pi(440) = 2764.60 \text{ rad/sec}$$

a) Position

$$x(t) = A \cos(\omega t) = 0.003 \cos(2764.60 t)$$

(phase = 0 as stated)

b) Velocity and Acceleration (and their maxima)

$$v(t) = -A\omega \sin(\omega t)$$

$$v_{\max} = A\omega = 0.003 \times 2764.60$$

$$v_{\max} = 8.2938 \text{ m/s}$$

$$a(t) = -A\omega^2 \cos(\omega t)$$

$$a_{\max} = A\omega^2 = 0.003 \times (2764.60)^2$$

$$a_{\max} = 2.2929 \times 10^4 \text{ m/s}^2$$

(c) Jerk (time derivative of acceleration)

$$j(t) = \frac{da}{dt} = A\omega^3 \sin(\omega t)$$

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$$j_{\max} = A\omega^3 = 0.003 \times (2764.60)^3$$

$$j_{\max} = 6.338 \times 10^7 \text{ m/s}^3$$

② $m = 2 \text{ kg}$, $K = 300 \text{ N/m}$

$$v_i = -12 \text{ m/s}, x_i = 0, t = 0$$

(a) :-

$$\omega = \sqrt{\frac{K}{m}} = \sqrt{\frac{300}{2}} = 12.25 \text{ rad/sec}$$

$$x(0) = A \cos(\phi) = 0$$

$$\cos(\phi) = 0$$

$$\phi = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$$

$$v = -A\omega \sin(\omega t + \phi)$$

$$v(0) = -A\omega \sin(\phi)$$

$$-12 = -A(12.25) \sin(\phi)$$

$$12 = 12.25 A(-1)$$

$$A = \frac{12}{12.25} = 0.98 \text{ m}$$

$$\phi = \frac{3\pi}{2} \text{ because } v \cos = -12 \rightarrow \text{negative}$$

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$$x(t) = 0.98 \sin(12.25t) \text{ m}$$

⑨ a) Frequency $f = \frac{1}{T}$

$$f = \frac{1}{10.0} = 0.100 \text{ Hz}$$

b)

$$A = 10.0 \text{ cm}$$

c)

$$T = 10.0 \text{ s}$$

d)

$$\omega = 2\pi f = 2\pi(0.1) = 0.2\pi \text{ rad/s}$$

$$\omega = 0.628 \text{ rad/sec}$$

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